Artifact Suppression and Contrast Enhancement in Reflection Ultrasound Computed Tomography

Soheil Hakakzadeh
Research and Development
Med Farasout Tajhiz Co.,
College of Engineering, University of Tehran,
Tehran, Iran
Soheilh@ieee.org

Zahra kavehvash
Research and Development
Med Farasout Tajhiz Co,.
Sharif University of Technology
Tehran, Iran
Kavehvash@sharif.edu

Mohammad Mehrmohammadi
Department of Imaging Science
University of Rochester Medical Center
Rochester, NY, USA
mohammad_mehr@urmc.rochester.edu

2025 IEEE International Ultrasonics Symposium (IUS) | 979-8-3315-2332-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/IUS62464.2025.11201801

***Abstract*— Reflection ultrasound computed tomography (RUCT) enables improved soft-tissue characterization but is often degraded by artifacts and limited contrast from sparse sampling and multipath reflections. We propose a statistical-consistency framework that reconstructs multiple low-quality images (LQIs) from transmitter subsets, computes pixel-wise correlations, and derives a stability ratio (SR) from the coefficient of variation (CV) to enhance stable structures. Experimental studies were performed on a published dataset [1] acquired with a 512-element transducer array (5 MHz, 60% bandwidth), including a cylindrical phantom and *in-vivo* mouse intestinal imaging. In the phantom study, our method improved contrast-to-noise ratio (CNR) by 11 dB and signal-to-noise ratio (SNR) by 12 dB, while *in-vivo* it achieved a 10 dB CNR and 8 dB SNR improvement. The results demonstrate that our method effectively suppresses artifacts and enhances structural visibility without hardware modification, providing a computationally efficient strategy for advancing RUCT in preclinical and clinical applications.**

***Keywords*— Reflection Ultrasound, Ultrasound Tomography, Image Reconstruction Algorithms, Artifact Suppression.**

I. Introduction

Reflection ultrasound computed tomography (RUCT) is a cutting-edge imaging technique that reconstructs maps of target's acoustic reflectivity. RUCT enables isotropic spatial resolution and improves soft-tissue characterization compared with conventional pulse-echo ultrasound. The ability to visualize structural features over a wide field of view without invasive procedures makes RUCT a promising modality for both clinical diagnostics and preclinical studies [2-4].

Despite these advantages, RUCT and other ultrasound-based imaging modalities remain constrained by image artifacts [5], limited contrast [6], and computational burden. Streaking [7], reverberations, sidelobes [8], and speckle-like interference commonly arise from sparse or irregular sampling, limited bandwidth [9], multipath propagation [10, 11], or heterogeneous speed-of-sound distributions [12-14]. Fundamental limitations such as spatial-domain pulse width broadening [15] and high computational cost of volumetric reconstructions [16, 17] further degrade image quality or restrict practical utility. On the systems side, the dense sampling requirements of circular transducer arrays require hundreds to thousands of elements, resulting in high hardware complexity, long scan time, and low patient throughput [1, 18]. Moreover, while micro-beamforming strategies [19, 20] are widely used in large transducer arrays to lower channel counts, their direct application to circular topologies remains challenging. Clinical implementation also faces challenges in acoustic coupling, motion stabilization, and standardization of imaging workflow [21]. These factors can obscure true tissue boundaries, reduce contrast, and degrade diagnostic reliability.

To address these issues, a wide range of algorithmic and system-level strategies have been proposed. Coherence-based methods such as the coherence factor (CF) have been incorporated into RUCT reconstructions to suppress incoherent clutter and improve resolution [13, 22, 23], or histogram-based method for methods for clutter reduction [24]. Adaptive beamforming approaches, including minimum variance (MV) and its multibeam implementations, have shown significant gains in lateral resolution and contrast using ring arrays [25, 26]. Transmission strategies have also evolved: diverging-wave excitation enables deeper penetration and a more uniform field of view [27], while plane-wave imaging combined with Eigen space-based MV beamforming reduces scan time without sacrificing resolution [26]. Synthetic aperture focusing techniques have been extended to multi-element or multi-mode configurations to optimize spatial resolution and SNR [28, 29], and have also been combined with structured illumination to achieve spatial resolution beyond the diffraction limit [30-32]. Beyond beamforming, signal-domain segmentation methods have been proposed to efficiently extract object boundaries prior to reconstruction [8,9]. Frequency-domain compressive imaging is also introduced for fast and data-efficient reconstruction [33]. More recently, machine learning has been leveraged to address data sparsity, with GAN-based and self-supervised frameworks demonstrating effective artifact suppression and contrast enhancement from limited transmissions [12, 34]. These advances highlight the growing interest in developing acquisition-efficient and computation-efficient strategies to address RUCT's practical and clinical limitations.

In parallel with these developments, we have previously introduced a family of artifact suppression and contrast-enhancement methods in photoacoustic computed tomography (PACT) based on statistical consistency analysis across multiple reconstructions [35, 36]. The core concept involves generating multiple low-quality images (LQIs) from subsets of the detected signals and quantifying their pixel-wise correlation to distinguish stable (true absorber) regions from unstable (artifact) regions. The resulting stability ratio is used to reweight the baseline reconstruction, enhancing consistent structures while attenuating artifacts [37, 38]. These studies demonstrated improved image fidelity and robustness in PACT across phantoms and *in-vivo* experiments.

Building on this prior body of work, we hypothesize that a statistical-consistency–based image enhancement framework, originally developed and validated for PACT, can be effectively adapted to RUCT to improve image contrast and structural stability without altering the acquisition hardware or protocol. To our knowledge, no prior study has applied such a framework to reflection-mode ultrasound tomography. This work addresses that gap by developing and evaluating an acquisition-agnostic, computationally efficient contrast-enhancement approach for RUCT, validated on both phantom and *in-vivo* data.

II. MATERIALS AND METHODS

In a ring-based RUCT system, a circular array of broadband transducers surrounds the target, sequentially transmitting and receiving acoustic pulses to reconstruct a two-dimensional map of acoustic reflectivity. For accurate image formation, adequate angular sampling is required to prevent aliasing and preserve spatial resolution. According to the Nyquist criterion, the angular sampling density must be at least twice the highest spatial frequency present in the reflected wavefield [39]. Sampling below this threshold increases the risk of arc-shaped streak artifacts, loss of contrast, and reduced image fidelity [40].

In a RUCT system, strong acoustic reflectors in the region of interest (ROI) are recorded by the array and reconstructed into an image using a back-projection or delay-and-sum (DAS) algorithm. Each transmitter emits the ultrasound wave into the ROI, and reflections are detected by all receivers, producing a dataset that can be reconstructed into a full-view baseline image. However, this process is susceptible to structured artifacts originating from reverberations, sidelobes, and multipath propagation.

Our proposed approach adapts the statistical-consistency framework previously developed for PACT to the reflection ultrasound domain [38]. In PACT, LQIs were formed by reconstructing data from varying subsets of receiver elements, exploiting the fact that limited-view artifacts change position across reconstructions while true absorbers remain spatially stable. In RUCT, we instead generate LQIs by partitioning the transmitter elements into at least three groups. Each group is used to acquire and reconstruct a separate image using the standard DAS algorithm. Varying the transmit aperture changes the field of view, resulting in differences in artifact position and intensity across LQIs, while genuine anatomical structures maintain consistent spatial characteristics.

The proposed technique's procedure is detailed below:

Step 1 – Initial image reconstruction:

Reconstruct the initial image from all transmitter–receiver pairs. Apply standard DAS reconstruction [1] to the entire dataset to produce the initial image.

Step 2 – Subsampled image reconstruction:

Partition the transmitter elements into no fewer than three groups, ensuring each group covers the ROI adequately. For each group, reconstruct an image independently using the DAS algorithm, producing multiple LQIs.

Step 3 – Pixel-wise correlation:

For each pixel within the ROI, compute the correlation between every pair of LQIs. This yields, for each pixel, a set of correlation coefficients representing the spatial stability of its intensity under varying transmit configurations.

Step 4 – Coefficient of variation (CV) calculation:

Calculate the CV of the pixel-wise correlation values for each pixel (x, y):

$$\mathrm{CV}(x, y) = \frac{\sigma_{\text{cross-corr}}(x,y)}{\mu_{\text{cross-corr}}(x,y)}, \tag{1}$$

where σ represents the standard deviation of pixel-wise correlation results, and μ denotes their average.

Step 5 – Stability ratio (SR) computation:

Invert the CV values and normalize them to the range [0,1], producing the SR map. True anatomical structures are represented by pixels with higher SR values, which show more consistent intensity across reconstructions.

Step 6 – SR-based image refinement:

To generate the final improved image, multiply the SR map to the initial image. This weighting improves contrast and overall image quality by enhancing stable features and suppressing areas that are prone to artifacts.

This statistical consistency-based improvement can be incorporated as a post-processing step in current RUCT reconstruction pipelines. The proposed method is completely acquisition-agnostic, and doesn't require any modifications to the scanning protocol or system hardware. Fig. 1 summarizes the entire workflow.

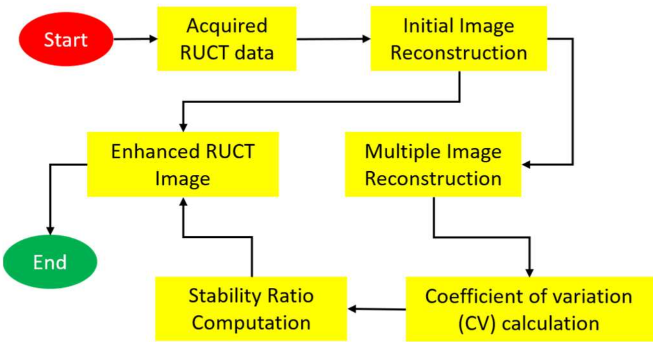

Fig. 1. Workflow for contrast enhancement in RUCT.

III. EXPERIMENTAL AND IN-VIVO RESULTS

A. Imaging System and Data Acquisition

The proposed method was evaluated using raw RUCT data obtained from a previously published study [1]. The acquisition was performed with a custom-built full-ring array (Imasonic SAS, Voray, France) with a 40-mm radius. The array comprised 512 elements arranged into two identical subarrays, each spanning 174° around the imaging target, with an interelement pitch of 0.47 mm. Each element measured 0.37 mm × 15 mm and was mounted on a concave surface to provide cylindrical focusing at 38 mm, near the array center.

The system operated at a center frequency of 5 MHz with a −6 dB fractional bandwidth of 60%, yielding an approximate beamwidth of 0.76 mm and a focal zone length of 13.5 mm per element. A custom data acquisition system (Falkenstein Mikrosysteme GmbH, Taufkirchen, Germany) delivered bipolar 38-Vpp transmit pulses at 5 MHz with programmable delays. In receive mode, the system digitized all channels simultaneously at up to 40 MSPS, transferring the acquired data to a host computer via 1-Gbit/s Ethernet.

B. Phantom Evaluation

A cylindrical agar phantom (20 mm diameter) with three 6-mm inclusions containing 50-µm microparticles was used to mimic reflective tissue structures [1]. Initial images were reconstructed with conventional DAS from the full dataset. LQIs were then generated by dividing the transmitter elements into three subsets, followed by pixel-wise correlation analysis, CV calculation, and SR derivation.

C. In-Vivo Assessment

In-vivo study was conducted on data from an eight-week-old female athymic nude mouse (Janvier Lab, France) under isoflurane anesthesia [1]. The animal was positioned upright in a water tank with anesthesia delivered via a breathing mask. Cross-sectional RUCT data were acquired from the intestinal region and processed with the same pipeline as in the phantom study (initial reconstruction, transmitter-subset LQIs, correlation analysis, CV calculation, SR derivation, and refinement).

D. Result Observations

Analysis of the phantom data indicated that the conventional DAS-based reconstruction method produced images with noticeable unwanted streak artifacts (see dashed circles and arrows in Fig. 2a). The proposed method, utilizing transmitter-subset LQIs and SR application, yielded sharper inclusion boundaries and substantially improved quantitative metrics. In particular, the SNR increased from 18 dB to 30 dB, and the CNR increased by 11 dB (Fig. 2b). Initial images for the *in-vivo* mouse experiment revealed hypoechoic organs that were partially hidden by artifacts and hyperechoic skin layers (Fig. 3a). By using the proposed technique, organ boundaries and vascular details were better defined, which increased the CNR by about 10 dB and the SNR from 14 dB to 22 dB (Fig. 3b).

These results demonstrate that the proposed method outperforms conventional RUCT reconstruction methods and offers significant contrast improvements in both phantom and *in-vivo* studies, which indicates a high degree of potential for translation to preclinical and clinical settings.

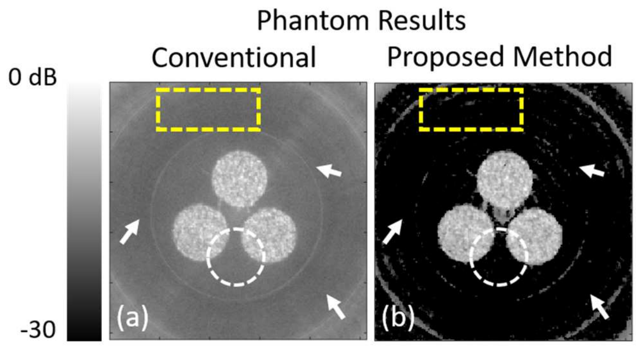

Fig. 2. Phantom images. (a) Conventional method, (b) SR-enhanced method with reduced artifacts.

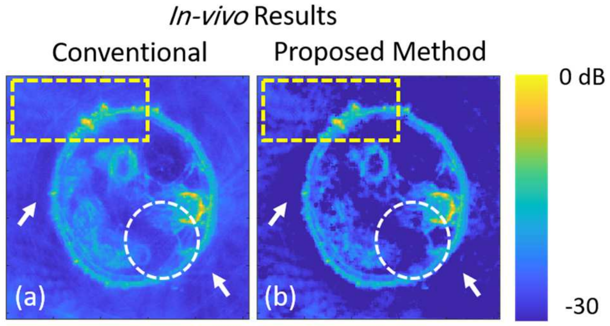

Fig. 3. *In-vivo* mouse intestinal cross-sections. (a) Conventional method, (b) SR-enhanced method with improved tissue contrast.

IV. Discussion and Conclusion

This study demonstrates that the proposed method can substantially improve image quality in a RUCT system by suppressing artifacts and enhancing contrast. Using both phantom and *in-vivo* data from a published dataset [1], we showed that incorporating transmitter-subset reconstructions and pixel-wise correlation provides a practical way to refine images without altering the acquisition hardware.

In the phantom experiments, our method suppresses streak artifacts and background level to an 11 dB increase in CNR and a 12 dB improvement in SNR compared with conventional DAS. The in-vivo mouse results showed clearer intestinal boundaries and vascular details, with a 10 dB gain in CNR and an 8 dB improvement in SNR. These improvements highlight our method's ability to maintain true tissue structures while suppressing artifacts from multipath reflections and sparse sampling.

A key strength of our method is its reliance on statistical consistency across transmitter-subset images, which makes it flexible and relatively easy to integrate into existing RUCT pipelines. However, the dependence on good initial reconstructions and the computational cost of correlation analysis should be evaluated in future work.

Overall, the SR-based enhancement method provides a computationally efficient way to improve image reliability. Future studies should focus on accelerating the pipeline for near–real-time use and testing the method across different anatomies and imaging conditions to further assess its translational potential.

References

[1] B. Lafci, J. Robin, X. L. Dean-Ben, and D. Razansky, "Expediting Image Acquisition in Reflection Ultrasound Computed Tomography," *IEEE Trans Ultrason Ferroelectr Freq Control,* vol. 69, no. 10, pp. 2837-2848, Oct 2022, doi: 10.1109/TUFFC.2022.3172713.

[2] P. J. Littrup, M. Mehrmohammadi, and N. Duric, "Breast Tomographic Ultrasound: The Spectrum from Current Dense Breast Cancer Screenings to Future Theranostic Treatments," *Tomography,* vol. 10, no. 4, pp. 554-573, Apr 15 2024, doi: 10.3390/tomography10040044.

[3] K. Kratkiewicz, A. Pattyn, N. Alijabbari, and M. Mehrmohammadi, "Ultrasound and Photoacoustic Imaging of Breast Cancer: Clinical Systems, Challenges, and Future Outlook," *J Clin Med,* vol. 11, no. 5, Feb 22 2022, doi: 10.3390/jcm11051165.

[4] N. V. Ruiter, M. Zapf, T. Hopp, and H. Gemmeke, "Ultrasound Tomography," *Adv Exp Med Biol,* vol. 1403, pp. 171-200, 2023.

[5] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvash, "Unipolar Back-Projection Algorithm for Photoacoustic Tomography," in *2022 IEEE*

International Ultrasonics Symposium (IUS), 10-13 Oct. 2022 2022, pp. 1-4, doi: 10.1109/IUS54386.2022.9957429.
[6] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvash, "Coherent Compounding for Photoacoustic Tomography," in *2023 IEEE International Ultrasonics Symposium (IUS)*, 3-8 Sept. 2023 2023, pp. 1-4, doi: 10.1109/IUS51837.2023.10307230.
[7] S. Hakakzadeh, P. Rajendran, V. A. Nili, Z. Kavehvash, and M. Pramanik, "A Spatial-Domain Factor for Sparse-Sampling Circular-View Photoacoustic Tomography," *IEEE Journal of Selected Topics in Quantum Electronics,* vol. 29, no. 4: Biophotonics, pp. 1-9, 2023.
[8] S. Hakakzadeh, Z. Kavehvash, and M. Mehrmohammadi, "Effective Apodization in Synthetic Aperture Ultrasound Computed Tomography," in *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, 2024: IEEE, pp. 1-4.
[9] S. Hakakzadeh, M. Amjadian, Y. Zhang, S. M. Mostafavi, Z. Kavehvash, and L. Wang, "Signal restoration algorithm for photoacoustic imaging systems," *Biomed Opt Express,* vol. 14, no. 2, pp. 651-666, Feb 1 2023, doi: 10.1364/BOE.480842.
[10] S. Hakakzadeh, Z. Kavehvash, and M. Mehrmohammadi, "Phased-Array Compounding for Reflection Ultrasound Computed Tomography Systems," in *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, 2024: IEEE, pp. 1-4.
[11] S. Hakakzadeh *et al.*, "Multi-angle data acquisition to compensate transducer finite size in photoacoustic tomography," *Photoacoustics,* vol. 27, p. 100373, 2022/09/01/ 2022.
[12] L. He *et al.*, "Self-Supervised Optimization of RF Data Coherence for Improving Breast Reflection UCT Reconstruction," *IEEE Trans Ultrason Ferroelectr Freq Control,* vol. 72, no. 8, pp. 1147-1159, Aug 2025, doi: 10.1109/TUFFC.2025.3581915.
[13] Z. Hou *et al.*, "Ultrasound Computed Tomography Reflection Imaging with Coherence-Factor Beamforming for Breast Tumor Early Detection," *Mathematics,* vol. 12, no. 7, p. 1106, 2024.
[14] C.-F. T. Chiang, I. Esteban, X.-L. Dean-Ben, J.-H. Hsu, D. Razansky, and H.-C. A. Lin, "Optimizing image fidelity in a spherical array geometry for pulse-echo ultrasound and volumetric optoacoustic imaging," *Photoacoustics,* vol. 45, 2025.
[15] Q. Zhang, Q. Liu, Z. Liu, M. Ding, and M. Yuchi, "Effect of spatial-domain pulse width on the resolution of scattering images in ultrasound computed tomography," *Phys Med Biol,* vol. 68, no. 17, Aug 14 2023.
[16] T. Liang, G. Jin, Y. Zeng, and X. Cai, "A signal-domain method to reconstruct object's boundary for 3D ultrasound computed tomography," presented at the 2023 IEEE International Ultrasonics Symposium (IUS), 2023.
[17] G. Jin *et al.*, "A Signal Domain Object Segmentation Method for Ultrasound and Photoacoustic Computed Tomography," *IEEE Trans Ultrason Ferroelectr Freq Control,* vol. PP, Dec 26 2022.
[18] L. Zhou *et al.*, "A coupling, stabilizing, and shaping strategy for breast ultrasound computed tomography (USCT) with a ring array transducer," *Ultrasonics,* vol. 138, p. 107212, Mar 2024, doi: 10.1016/j.ultras.2023.107212.
[19] V. A. Nili, Y. Yan, M. Ezati, Z. Kavehvash, D. Rubens, and M. Mehrmohammadi, "Improved Subarray Ultrasound Beamforming Using a Spatial-Frequency Domain Analysis," *IEEE Transactions on Instrumentation and Measurement,* vol. 74, pp. 1-15, 2025.
[20] V. A. Nili, M. Ezati, Y. Yan, Z. Kavehvash, and M. Mehrmohammadi, "Field of View and Resolution Improvement in Coprime Sparse Synthetic Aperture Ultrasound Imaging," in *2022 IEEE International Ultrasonics Symposium (IUS)*, 10-13 Oct. 2022 2022, pp. 1-4.
[21] N. Ruiter, M. Zapf, R. Dapp, T. Hopp, W. Kaiser, and H. Gemmeke, "First results of a clinical study with 3D ultrasound computer tomography," in *2013 IEEE international ultrasonics symposium (IUS)*, 2013: IEEE, pp. 651-654.
[22] Z. Lan, C. Rong, C. Han, X. Qu, J. Li, and H. Lin, "A joint method of coherence factor and nonlinear beamforming for synthetic aperture imaging with a ring array," in *2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2023: IEEE, pp. 1-4.
[23] S. Hakakzadeh and Z. Kavehvash, "A Hilbert-based Coherence Factor for Photoacoustic Imaging," in *2022 29th National and 7th International Iranian Conference on Biomedical Engineering (ICBME)*, 21-22 Dec. 2022 2022, pp. 172-176, doi: 10.1109/ICBME57741.2022.10052999.
[24] V. A. Nili, M. Fakharzadeh, and Z. Kavehvash, "Clutter rejection and object separation in active milimeter-wave imaging system," in *2016 IEEE International Symposium on Antennas and Propagation (APSURSI)*, 26 June-1 July 2016 2016, pp. 177-178.
[25] H. Zhang, Y. Zhu, B. Zhou, S. Liu, N. He, and C. Tian, "Multibeam minimum variance beamforming for ring array ultrasound imaging," *Phys Med Biol,* vol. 68, no. 13, Jun 23 2023
[26] X. Jiang, Y. Xiao, Y. Wang, J. Yu, and H. Zheng, "Plane wave imaging combined with eigenspace-based minimum variance beamforming using a ring array in ultrasound computed tomography," *Biomed Eng Online,* vol. 18, no. 1, p. 7, Jan 23 2019.
[27] G. Jin, Y. Yan, and M. Mehrmohammadi, "Improving Reflection Mode Ultrasound Computed Tomography Using Diverging Waves," presented at the 2024 IEEE UFFC Latin America Ultrasonics Symposium (LAUS), 2024.
[28] L. Shi, H. Wang, X. Huang, X. Yuan, Q. Ding, and W. Zhang, "Ultrasound computed tomography image reconstruction with multi-mode aperture matching of ring array," *Measurement Science and Technology,* vol. 35, no. 3, 2023, doi: 10.1088/1361-6501/ad1579.
[29] Y. Wu *et al.*, "Research on the Multi-Element Synthetic Aperture Focusing Technique in Breast Ultrasound Imaging, Based on the Ring Array," *Micromachines (Basel),* vol. 13, no. 10, Oct 16 2022.
[30] V. A. Nili, Z. Kavehvash, S. Mostafavi, M. Shabany, and M. Mehrmohammadi, "Structured Illumination for Super-Resolution Synthetic Aperture Ultrasound Imaging," *IEEE Sensors Journal,* 2025.
[31] V. A. Nili, Y. Yan, S. Hakakzadeh, Z. Kavehvash, and M. Mehrmohammadi, "A Strategy for Structured Illumination in Synthetic Aperture Ultrasound Imaging," in *2023 IEEE International Ultrasonics Symposium (IUS)*, 3-8 Sept. 2023 2023, pp. 1-4.
[32] V. A. Nili, Z. Kavehvash, and M. Mehrmohammadi, "Structured Illumination and Plane-Wave Compounding: A Synergistic Approach to Super-Resolution Ultrasound," in *IEEE International Ultrasonics Symposium (IUS)*, 2025.
[33] A. Ghanbarzadeh-Dagheyan *et al.*, "Time-domain ultrasound as prior information for frequency-domain compressive ultrasound for intravascular cell detection: A 2-cell numerical model," *Ultrasonics,* vol. 125, p. 106791, 2022/09/01/ 2022.
[34] Z. Liu *et al.*, "Reconstruction of reflection ultrasound computed tomography with sparse transmissions using conditional generative adversarial network," *Ultrasonics,* vol. 145, p. 107486, Jan 2025.
[35] S. Hakakzadeh, Z. Kavehvash, and M. Pramanik, "Artifact Removal Factor for Circular-view Photoacoustic Tomography," in *2022 IEEE International Ultrasonics Symposium (IUS)*, 10-13 Oct. 2022 2022, pp. 1-4, doi: 10.1109/IUS54386.2022.9958228.
[36] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvash, "Fast Adapted Delay and Sum Reconstruction Algorithm in Circular Photoacoustic Tomography," in *2022 30th International Conference on Electrical Engineering (ICEE)*, 17-19 May 2022 2022, pp. 49-54.
[37] S. Hakakzadeh, P. Rajendran, Z. Kavehvash, and M. Pramanik, "Enhancing image quality in circular-view photoacoustic tomography using randomized detection points," *Journal of Physics: Photonics,* vol. 7, no. 1, p. 015006, 2024.
[38] S. Hakakzadeh, Z. Kavehvash, and M. Mehrmohammadi, "Enhancing Contrast in Circular-view Photoacoustic Computed Tomography Systems," in *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, 2024: IEEE, pp. 1-4.
[39] V. Amin Nili, E. Mansouri, Z. Kavehvash, M. Fakharzadeh, M. Shabany, and A. Khavasi, "Low-cost three-dimensional millimeter-wave holographic imaging system based on a frequency-scanning antenna," *Applied Optics,* vol. 57, no. 1, pp. A65-A75, 2018.
[40] P. Hu, L. Li, and L. V. Wang, "Location-Dependent Spatiotemporal Antialiasing in Photoacoustic Computed Tomography," *IEEE Trans Med Imaging,* vol. 42, no. 4, pp. 1210-1224, Apr 2023.